

# Chapter 1: Complex Analysis

## Mathematical Physics

Compiled by Jake W. Liu

September 17, 2026

- 1 Taylor series representation
- 2 Complex variables
- 3 Derivatives
- 4 Path integrals
- 5 Taylor and Laurent series
- 6 Singularities
- 7 Residues
- 8 Residue calculus
- 9 Method of steepest descent

# Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (complex differentiable) obey the *Cauchy–Riemann equations* and admit power-series expansions.
- **Cauchy’s integral theorem** and the **residue theorem** turn contour integrals into a sum of local residues—the tool used later for Gamma/Beta identities and real integrals.
- A short catalog of **contour templates** (semicircle, keyhole, Jordan) shows how orientation and vanishing-arc estimates convert complex residues into real answers.
- The **method of steepest descent** extracts large-parameter asymptotics of contour integrals. Its real special cases (Laplace method, stationary phase) reappear for Stirling (Ch. 2) and for large-argument Bessel expansions (Ch. 5).

# Matching a polynomial to a function

A Taylor polynomial replaces a function near one point by a polynomial with the same value, slope, curvature, and higher derivatives there.

Start at  $x = 0$  and write the first few terms rather than a general sum:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_Nx^N \quad (N \geq 3).$$

Differentiate and then set  $x = 0$ :

$$p(0) = a_0,$$

$$p'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots,$$

$$p'(0) = a_1,$$

$$p''(x) = 2a_2 + 6a_3x + \cdots,$$

$$p''(0) = 2!a_2,$$

$$p^{(3)}(x) = 6a_3 + \cdots,$$

$$p^{(3)}(0) = 3!a_3.$$

To match  $f$ , choose  $a_0 = f(0)$ ,  $a_1 = f'(0)$ ,  $a_2 = f''(0)/2!$ , and so on. The pattern is

$$a_k = \frac{f^{(k)}(0)}{k!}. \quad (1.1.1)$$

The factorial appears because differentiating  $x^k$  exactly  $k$  times gives  $k!$ .

# Matching a polynomial to a function (cont.)

At a different center  $x_0$ , use powers of the distance  $x - x_0$ :

$$T_N(x; x_0) = \sum_{k=0}^N \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.2)$$

For example, the quadratic approximation is

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2.$$

**How accurate is it?** For a real-valued  $f$  and real  $x \neq x_0$ , assume  $f$  has  $N + 1$  continuous derivatives between  $x_0$  and  $x$ , Taylor's theorem gives

$$f(x) = T_N(x; x_0) + R_N(x), \quad R_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x_0)^{N+1}, \quad (1.1.3)$$

where  $\xi$  lies between  $x_0$  and  $x$ . At  $x = x_0$  the error is zero. Thus a bound on the next derivative gives a bound on the error.

When  $R_N(x) \rightarrow 0$ , the polynomial becomes an exact infinite series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.4)$$

Having derivatives of every order is not, by itself, enough to guarantee this last equality.

# Examples: the elementary power series

**Exponential.** Every derivative of  $e^x$  equals  $e^x$ , so every derivative at zero equals 1. Therefore

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots. \quad (1.1.5)$$

For fixed  $x \neq 0$ , the ratio of consecutive absolute terms is

$$\frac{|x|^{k+1}/(k+1)!}{|x|^k/k!} = \frac{|x|}{k+1} \rightarrow 0.$$

The series converges for every  $x$ . For real  $x$ , it represents  $e^x$  because

$$|R_N(x)| \leq e^{|x|} \frac{|x|^{N+1}}{(N+1)!} \rightarrow 0.$$

For example, at  $x = 0.1$  the first three terms give  $e^{0.1} \approx 1 + 0.1 + 0.005 = 1.105$ .

# Examples: the elementary power series (cont.)

**Cosine and sine.** List the derivative values at zero and divide by the corresponding factorial:

| $k$               | 0 | 1 | 2  | 3  | 4 | 5 |
|-------------------|---|---|----|----|---|---|
| $(\cos)^{(k)}(0)$ | 1 | 0 | -1 | 0  | 1 | 0 |
| $(\sin)^{(k)}(0)$ | 0 | 1 | 0  | -1 | 0 | 1 |

Only even powers survive for cosine; only odd powers survive for sine:

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad (1.1.6)$$

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots. \quad (1.1.7)$$

For real  $x$ , their Taylor errors obey  $|R_N(x)| \leq |x|^{N+1}/(N+1)! \rightarrow 0$  because all derivatives have magnitude at most 1. All three series also converge for every complex argument. We use them to extend  $e^x$ ,  $\cos x$ , and  $\sin x$  to complex numbers.

# Imaginary unit, Cartesian and polar form

Define the imaginary unit  $i$  by  $i^2 = -1$ ; the other square root of  $-1$  is  $-i$ . A complex number has Cartesian form

$$z = x + iy, \quad x = \operatorname{Re} z, \quad y = \operatorname{Im} z,$$

where  $x, y \in \mathbb{R}$ ;  $\operatorname{Re} z$  and  $\operatorname{Im} z$  denote the real and imaginary parts of  $z$ . The powers of  $i$  repeat with period four:

$$i^{4q} = 1, \quad i^{4q+1} = i, \quad i^{4q+2} = -1, \quad i^{4q+3} = -i.$$

Substitute  $i\theta$  into the absolutely convergent exponential series and separate even and odd powers:

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{i^{2k} \theta^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{i^{2k+1} \theta^{2k+1}}{(2k+1)!} \end{aligned}$$

and since  $i^{2k} = (i^2)^k = (-1)^k$  and  $i^{2k+1} = i i^{2k} = i(-1)^k$ ,

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!}.$$

# Imaginary unit, Cartesian and polar form (cont.)

Comparison with (1.1.6)–(1.1.7) gives Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (1.2.1)$$

The exponential addition law also holds for complex arguments:

$$e^{w_1+w_2} = e^{w_1} e^{w_2}.$$

It follows by multiplying the two absolutely convergent series and applying the binomial theorem. In particular,

$$e^{u+iv} = e^u (\cos v + i \sin v), \quad u, v \in \mathbb{R}. \quad (1.2.2)$$

The factor  $e^u$  determines the modulus, and  $v$  determines the angle.

For  $z \neq 0$ , set

$$r = |z| = \sqrt{x^2 + y^2} > 0.$$

Choose any angle  $\theta$  satisfying  $x = r \cos \theta$  and  $y = r \sin \theta$ . Then

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}. \quad (1.2.3)$$

The angle is not unique: if  $\theta$  is one argument, then every argument is  $\theta + 2\pi k$ ,  $k \in \mathbb{Z}$ . The principal argument is commonly chosen as  $\text{Arg } z \in (-\pi, \pi]$  (other endpoint conventions are also used). The number  $z = 0$  has modulus 0 but no defined argument.

# Imaginary unit, Cartesian and polar form (cont.)

If  $z_1 = r_1 e^{i\theta_1}$  and  $z_2 = r_2 e^{i\theta_2}$ , then the phases add by the addition law above, so

$$z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)},$$

so moduli multiply and arguments add modulo  $2\pi$ .

The complex conjugate is

$$\bar{z} = x - iy = re^{-i\theta}.$$

Consequently

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = r^2 = |z|^2,$$

and adding or subtracting  $z$  and  $\bar{z}$  isolates the Cartesian components:

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}. \quad (1.2.4)$$

Replacing  $\theta$  by  $-\theta$  in Euler's formula gives  $e^{-i\theta} = \cos \theta - i \sin \theta$ . Hence

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta,$$

and therefore

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (1.2.5)$$

# Multivalued functions and branch cuts

An inverse relation need not define a single-valued function. For example,  $y = x^2$  gives  $x = \pm\sqrt{y}$  for  $y > 0$ . A *branch* selects one value continuously on a chosen domain; the usual real principal square root selects  $+\sqrt{y}$  for  $y \geq 0$ .

The complex logarithm has the same issue: several inputs have the same exponential. **The complex logarithm.** Let  $z \neq 0$  and write  $z = re^{i\theta}$  with  $r > 0$ . To solve  $e^w = z$ , write  $w = u + iv$  with  $u, v \in \mathbb{R}$ . The splitting (1.2.2) gives

$$e^w = e^{u+iv} = e^u(\cos v + i \sin v).$$

Equality with  $r(\cos \theta + i \sin \theta)$  requires

$$e^u = r \implies u = \log r,$$

and

$$\cos v = \cos \theta, \quad \sin v = \sin \theta \implies v = \theta + 2\pi k, \quad k \in \mathbb{Z}.$$

Thus the complete multivalued logarithm is

$$\log z = \log |z| + i(\theta + 2\pi k), \quad k \in \mathbb{Z}. \quad (1.2.6)$$

A principal branch is obtained by choosing one interval of length  $2\pi$ , commonly

$$-\pi < \operatorname{Arg} z < \pi, \quad \operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

This analytic branch is defined on  $\mathbb{C} \setminus (-\infty, 0]$ . The values  $\pm\pi$  are limits from the two sides of the cut, not values on this domain.

# Multivalued functions and branch cuts (cont.)

A continuous argument increases by  $2\pi$  after one counterclockwise turn around 0. Thus the logarithm does not return to its starting value: 0 is a *branch point*. Removing a *branch cut* prevents such a loop from staying in our domain. For the principal logarithm, the cut is the nonpositive real axis. Its two boundary values are

$$\operatorname{Log}(-r + i0) = \log r + i\pi, \quad \operatorname{Log}(-r - i0) = \log r - i\pi,$$

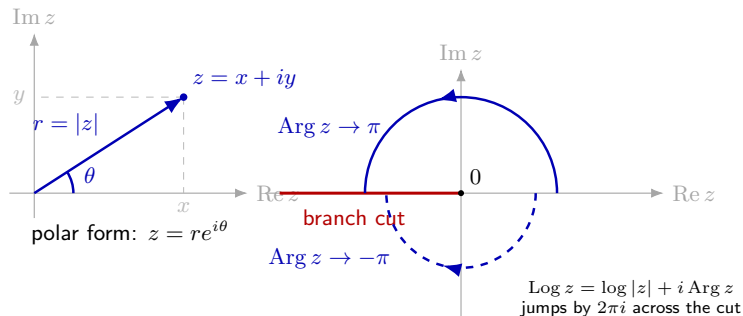
so crossing the cut changes the logarithm by  $2\pi i$ .

The same mechanism applies to roots. For example,

$$z^{1/2} = \sqrt{r} e^{i(\theta+2\pi k)/2} = (-1)^k \sqrt{r} e^{i\theta/2},$$

so there are two values. A branch cut permits one of them to be selected continuously.

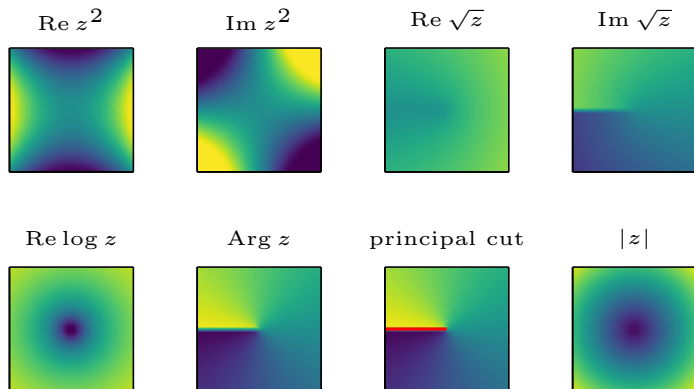
# Pictures: branch cuts and component plots



Across the removed ray, the two boundary values of the chosen argument differ by  $2\pi$ . The branch is defined on the rest of the cut plane.

# Pictures: branch cuts and component plots (cont.)

A complex map  $z = x + iy \mapsto f = u + iv$  has two real inputs and two real outputs, so we use cut planes and component plots rather than a single graph  $y = f(x)$ .



# The complex derivative and Cauchy–Riemann

The complex derivative at  $z_0$  is

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad (1.3.1)$$

The increment  $h$  may approach zero from any direction in the complex plane. The same limit must result in every case. A function is *holomorphic* (analytic) on an open set if this derivative exists at every point there; it is *entire* if that set is all of  $\mathbb{C}$ .

Write  $z = x + iy$  and  $f(z) = u(x, y) + iv(x, y)$ . Here  $u_x = \partial u / \partial x$  means differentiate in  $x$  while holding  $y$  fixed;  $u_y$ ,  $v_x$ , and  $v_y$  are defined in the same way.

Along a real increment  $h = \Delta x$ , the difference quotient is

$$\begin{aligned} \frac{f(z + h) - f(z)}{h} &= \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\ &\xrightarrow{\Delta x \rightarrow 0} u_x + iv_x, \end{aligned}$$

so  $f'(z) = u_x + iv_x$ . Along an imaginary increment  $h = i\Delta y$  the bookkeeping is the same, with the extra factor  $1/i$  pulled out of the denominator:

$$\frac{f(z + i\Delta y) - f(z)}{i\Delta y} = \frac{1}{i} \left[ \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \right].$$

# The complex derivative and Cauchy–Riemann (cont.)

If the complex derivative exists, both directional calculations must give that derivative. Passing to the limit and using  $1/i = -i$ ,

$$f'(z) = \frac{u_y + iv_y}{i} = -i(u_y + iv_y) = v_y - iu_y.$$

Because  $f$  is complex differentiable, every direction of approach must return the *same* value  $f'(z)$ ; equating  $u_x + iv_x = v_y - iu_y$  and matching real and imaginary parts gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x. \quad (1.3.2)$$

They are necessary for complex differentiability.

**Working test.** If  $u$  and  $v$  have continuous first partial derivatives near a point, the Cauchy–Riemann equations are also sufficient there, and

$$f'(z) = u_x + iv_x = v_y - iu_y.$$

Thus our procedure is: find  $u, v$ , compute four partial derivatives, and compare.

**Example:**  $z^2$ . Since

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

we have

$$u_x = 2x = v_y, \quad u_y = -2y = -v_x.$$

Thus  $z^2$  is entire and  $f'(z) = 2x + i(2y) = 2z$ .

# The complex derivative and Cauchy–Riemann (cont.)

**Counterexample:**  $\bar{z} = x - iy$ . Then  $u_x = 1 \neq -1 = v_y$ , so CR fails everywhere. Equivalently  $\bar{h}/h$  is 1 for real  $h$  and  $-1$  for imaginary  $h$ : no direction-independent derivative.

Keep in mind that an analytic function is infinitely differentiable and equals its Taylor series on every disk where it is analytic; we prove this once contour integrals are available.

# Contour integrals and a key circle integral

A contour integral is an ordinary integral along a chosen path. Parameterize the path by  $z = \gamma(t)$ ,  $a \leq t \leq b$ , with  $\gamma$  piecewise continuously differentiable. Since  $dz = \gamma'(t) dt$ ,

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt. \quad (1.4.1)$$

Reversing the path reverses the sign:

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz.$$

A closed contour may be traversed in either direction. A positively oriented boundary keeps the region on the left: the outer boundary is counterclockwise, while a boundary around a hole is clockwise.

**Integer powers on a circle.** Let  $n \in \mathbb{Z}$  and parameterize the positively oriented circle  $|z| = R$  by

$$z(\theta) = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta, \quad 0 \leq \theta \leq 2\pi.$$

# Contour integrals and a key circle integral (cont.)

Then

$$\begin{aligned}\oint_{|z|=R} z^n dz &= \int_0^{2\pi} R^n e^{in\theta} (iRe^{i\theta}) d\theta \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta.\end{aligned}$$

If  $n = -1$ , the exponential is 1, so

$$\oint_{|z|=R} \frac{dz}{z} = i \int_0^{2\pi} d\theta = 2\pi i.$$

If  $n \neq -1$ , then

$$\begin{aligned}iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta &= iR^{n+1} \left[ \frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{R^{n+1}}{n+1} (e^{2\pi i(n+1)} - 1) = 0,\end{aligned}$$

because  $n+1$  is an integer. Let  $\delta_{n,-1}$  denote the Kronecker delta: it is 1 when  $n = -1$  and 0 otherwise. Translating the circle gives

$$\oint_{|z-z_0|=R} (z-z_0)^n dz = 2\pi i \delta_{n,-1}. \quad (1.4.2)$$

# Cauchy's integral theorem and formula

A *domain* is an open, connected set. A domain is *simply connected* if every closed curve in it can be continuously contracted to a point without leaving the domain; for the planar regions used here, this is the precise version of “the region has no holes.”

## Cauchy's integral theorem

Let  $D$  be a simply connected domain, let  $f$  be holomorphic on  $D$ , and let  $\gamma$  be a closed piecewise smooth contour in  $D$ . Then

$$\oint_{\gamma} f(z) dz = 0.$$

**Why zero?** A direct consequence of Green's theorem and the Cauchy–Riemann equations. With  $f = u + iv$  and  $dz = dx + i dy$ ,

$$\begin{aligned}\oint_{\gamma} f dz &= \oint_{\gamma} (u dx - v dy) + i \oint_{\gamma} (v dx + u dy) \\ &= \iint_E (-v_x - u_y) dA + i \iint_E (u_x - v_y) dA \\ &= \iint_E (u_y - u_y) dA + i \iint_E (v_y - v_y) dA = 0.\end{aligned}$$

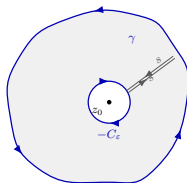
# Cauchy's integral theorem and formula (cont.)

## Cauchy's integral formula

Let  $\gamma$  be a positively oriented simple closed contour, let  $f$  be holomorphic on an open set containing  $\gamma$  and its interior, and let  $z_0$  lie inside  $\gamma$ . Then

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz. \quad (1.4.3)$$

To understand the formula, deform  $\gamma$  to a small counterclockwise circle  $C_\varepsilon$  about  $z_0$ . The integrand is holomorphic between the two curves, so their integrals agree. The two sides of the connecting slit cancel.



Slit construction: the combined contour  $\gamma + s - C_\varepsilon - s$  encloses the shaded region, where the integrand is holomorphic; the inner circle is traversed clockwise and the slit sides  $s$  and  $-s$  cancel.

# Cauchy's integral theorem and formula (cont.)

On the small circle  $z = z_0 + \varepsilon e^{i\theta}$ ,

$$\oint_{C_\varepsilon} \frac{f(z)}{z - z_0} dz = i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta.$$

As  $\varepsilon \rightarrow 0$ , continuity makes the integrand approach  $f(z_0)$ , so the last integral approaches  $2\pi i f(z_0)$ . This gives (1.4.3).

Differentiate (1.4.3) in  $z_0$  (contour fixed, away from  $z_0$ ). Since  $\partial_{z_0}^n (z - z_0)^{-1} = n! / (z - z_0)^{n+1}$ ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (1.4.4)$$

In particular every holomorphic function is infinitely differentiable, and

$$\oint_\gamma \frac{f(z)}{(z - z_0)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(z_0).$$

# Taylor series from contour integration

## Taylor series theorem

If  $f$  is holomorphic on the disk  $|z - z_0| < R$ , then for every  $|z - z_0| < R$ ,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Choose a radius  $\rho$  satisfying

$$|z - z_0| < \rho < R$$

and integrate on  $C_\rho : |\xi - z_0| = \rho$ . Cauchy's formula gives

$$f(z) = \frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{\xi - z} d\xi.$$

The idea is to expand only the kernel, using the geometric series:

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \frac{1}{1 - (z - z_0)/(\xi - z_0)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}.$$

# Taylor series from contour integration (cont.)

This works because  $|(z - z_0)/(\xi - z_0)| = |z - z_0|/\rho < 1$  all along the circle. Substituting and integrating the convergent series term by term gives

$$f(z) = \sum_{n=0}^{\infty} (z - z_0)^n \underbrace{\frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi}_{f^{(n)}(z_0)/n! \text{ by (1.4.4)}}.$$

The coefficients are exactly those found by matching derivatives at the start of the chapter. Here Cauchy's formula also tells us where the series equals  $f$ .

The key consequence is simple: inside a disk where  $f$  is holomorphic, its Taylor series represents  $f$ . For an expansion about  $z_0$ , the nearest nonremovable singularity limits the radius of convergence. Boundary points require separate checking.

The converse also holds: a convergent power series may be differentiated term by term inside its disk, so it is holomorphic there. Being analytic at  $z_0$  and having a Taylor series representation about  $z_0$  are the same property.

# Analytic continuation

A Taylor-series expansion is local: it may stop converging even though the function itself remains holomorphic farther away.

For example,

$$f(z) = \frac{1}{1-z} = 1 + z + z^2 + \cdots, \quad |z| < 1.$$

The Taylor series about  $z = 0$  converges only for  $|z| < 1$ , because the nearest singularity is at  $z = 1$ . However, the function

$$f(z) = \frac{1}{1-z}$$

is still holomorphic at every point except  $z = 1$ .

To go beyond the original Taylor disk, choose a new center inside the region where  $f$  is already known. For example, about  $z = -\frac{1}{2}$ ,

$$\begin{aligned} f(z) &= \frac{1}{\frac{3}{2} - (z + \frac{1}{2})} \\ &= \frac{2}{3} \frac{1}{1 - \frac{2}{3}(z + \frac{1}{2})} \\ &= \frac{2}{3} \sum_{n=0}^{\infty} \left[ \frac{2}{3} \left( z + \frac{1}{2} \right) \right]^n, \end{aligned}$$

# Analytic continuation (cont.)

which converges for

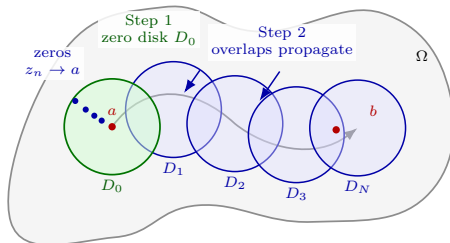
$$\left| z + \frac{1}{2} \right| < \frac{3}{2}.$$

This new Taylor disk overlaps the original disk but reaches farther.

More generally, we may repeat this process:

$$D_0 \longrightarrow D_1 \longrightarrow D_2 \longrightarrow \dots$$

On every overlap, the Taylor expansions represent the same holomorphic function and therefore must agree. In this way, the function can be continued step by step through a larger connected domain.



# Laurent series

A Taylor series uses nonnegative powers. Near a singularity we also need  $1/(z - z_0), 1/(z - z_0)^2, \dots$ . Their combination is a *Laurent series*.

Suppose  $f$  is holomorphic in the annulus

$$r < |z - z_0| < R.$$

An inner hole allows negative powers; the next singularity limits the outer radius.  $f$  has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \underbrace{\sum_{n=0}^{\infty} a_n(z - z_0)^n}_{\text{regular part}} + \underbrace{\sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}}_{\text{principal part}}, \quad (1.5.1)$$

convergent throughout the annulus. To extract a coefficient, multiply by  $(z - z_0)^{-k-1}$  and integrate on a centered circle  $\gamma : |z - z_0| = \rho$ , where  $r < \rho < R$ , oriented counterclockwise:

$$\begin{aligned} \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz &= \frac{1}{2\pi i} \sum_{n=-\infty}^{\infty} a_n \oint_{\gamma} (z - z_0)^{n-k-1} dz \\ &= \frac{1}{2\pi i} a_k (2\pi i) = a_k. \end{aligned}$$

## Laurent series (cont.)

Therefore, for every integer  $k$ ,

$$a_k = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz. \quad (1.5.2)$$

On the chosen circle inside the annulus, term-by-term integration is valid.

If the principal part is empty, the series extends holomorphically to the full disk and is a Taylor series. A punctured disk  $0 < |z - z_0| < R$  does not guarantee this:  $1/(z - z_0)$  is the simplest counterexample.

# Classifying singularities

A point  $z_0$  is a *singular point* of  $f$  when  $f$  fails to be holomorphic there. It is an *isolated singularity* if there exists  $R > 0$  such that  $f$  is holomorphic on the punctured disk

$$0 < |z - z_0| < R.$$

Only isolated singularities admit an ordinary Laurent classification. Write

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Then:

- The singularity is **removable** if  $a_n = 0$  for every  $n < 0$ . Defining  $f(z_0) = a_0$  makes  $f$  holomorphic at  $z_0$ . Example:  $\sin z/z$  at 0.
- It is a **pole of order  $m$** , where  $m \geq 1$  is an integer, if  $a_{-m} \neq 0$  and  $a_n = 0$  for every  $n < -m$ . Equivalently,

$$f(z) = \frac{h(z)}{(z - z_0)^m}, \quad h \text{ holomorphic, } h(z_0) \neq 0.$$

Multiplying by  $(z - z_0)^m$  removes exactly the highest negative power.

# Classifying singularities (cont.)

- It is **essential** if infinitely many negative-power coefficients are nonzero. For example,

$$\sin\left(\frac{1}{z}\right) = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \cdots$$

has an essential singularity at 0.

A function holomorphic except for isolated poles is *meromorphic*. For example,  $1/\sin z$  is meromorphic, with simple poles at  $z = n\pi$ ,  $n \in \mathbb{Z}$ .

To check this example,  $\sin z = 0$  is equivalent to  $e^{2iz} = 1$ . Writing  $z = x + iy$ , its modulus gives  $e^{-2y} = 1$ , so  $y = 0$ ; its phase then gives  $x = n\pi$ . Near such a point, Taylor expansion gives

$$\sin z = \cos(n\pi)(z - n\pi) + \cdots = (-1)^n(z - n\pi) + \cdots.$$

Thus  $1/\sin z$  begins with  $(-1)^n/(z - n\pi)$  and has a simple pole there.

A branch point is not an isolated singularity. For example, analytic continuation of  $\log z$  once around 0 changes its value by  $2\pi i$ , so no single-valued branch is holomorphic on a complete punctured disk around 0. The removable/pole/essential trichotomy therefore does not apply to the branch point itself.

At infinity, inspect  $F(w) = f(1/w)$  at  $w = 0$ . A polynomial of degree  $m \geq 1$  has a pole of order  $m$  there; a constant is removable;  $e^z$  is essential.

# The residue theorem

For an isolated singularity  $z_0$ , the residue is the Laurent coefficient

$$\operatorname{Res}(f, z_0) = a_{-1}.$$

Choose a small counterclockwise circle  $C$  about  $z_0$  enclosing no other singularity. Use the Laurent expansion valid near  $z_0$ , not one on an annulus containing other singularities. By (1.4.2), integrating any integer power around  $C$  gives zero except for  $(z - z_0)^{-1}$ , whose integral is  $2\pi i$ . Only the  $n = -1$  term survives:

$$\begin{aligned}\oint_C f(z) dz &= \sum_{n=-\infty}^{\infty} a_n \oint_C (z - z_0)^n dz \\ &= a_{-1}(2\pi i) \\ &= 2\pi i \operatorname{Res}(f, z_0).\end{aligned}$$

Thus

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}(f, z_0). \quad (1.7.1)$$

# The residue theorem (cont.)

## Residue theorem

Let  $f$  be holomorphic inside and on a positively oriented simple closed contour  $\gamma$ , except for finitely many isolated singularities  $z_1, \dots, z_N$  in the interior. Then

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^N \operatorname{Res}(f, z_k). \quad (1.7.2)$$

**Geometric reason.** Deform the outer contour into small counterclockwise circles around the singularities. Each circle contributes its own  $2\pi i$  times residue. The simple outer contour encloses each of these singularities once.

# The residue theorem (cont.)

**Residue at a pole of order  $m$ .** Write

$$h(z) = (z - z_0)^m f(z),$$

where  $h$  is holomorphic and  $h(z_0) \neq 0$ . Taylor expansion gives

$$h(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^j.$$

Dividing by  $(z - z_0)^m$ ,

$$f(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^{j-m}.$$

The exponent equals  $-1$  exactly when  $j = m - 1$ . Therefore

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \bigg|_{z=z_0}. \quad (1.7.3)$$

For  $m = 1$  this reduces to

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z),$$

and for  $m = 2$  it becomes the first derivative of  $(z - z_0)^2 f(z)$  at  $z_0$ .

# The residue theorem (cont.)

**A quotient at a simple zero.** If  $q(z_0) = 0$  and  $q'(z_0) \neq 0$ , then

$$q(z) = q'(z_0)(z - z_0) + \cdots.$$

For  $p, q$  holomorphic near  $z_0$ , cancel  $z - z_0$  before taking the limit:

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \lim_{z \rightarrow z_0} \frac{p(z)}{q(z)/(z - z_0)} = \frac{p(z_0)}{q'(z_0)}. \quad (1.7.4)$$

When  $p(z_0) = 0$ , the singularity is removable and the formula gives zero.

For example,

$$\operatorname{Res}\left(\frac{1}{z^2 + 1}, i\right) = \frac{1}{2i},$$

while the order-2 formula at the double pole of  $1/(z^2 + 1)^2$  gives

$$\operatorname{Res}\left(\frac{1}{(z^2 + 1)^2}, i\right) = \left. \frac{d}{dz} \frac{1}{(z + i)^2} \right|_{z=i} = -\frac{2}{(2i)^3} = \frac{1}{4i}.$$

# The residue theorem (cont.)

**Quick checks on  $|z| = 1$ .**

$$\frac{\sin z}{z^2} = \frac{1}{z} - \frac{z}{3!} + \cdots \implies \operatorname{Res}\left(\frac{\sin z}{z^2}, 0\right) = 1,$$

so  $\oint_{|z|=1} (\sin z)/z^2 dz = 2\pi i$ . By contrast  $z^2/\sin z$  extends holomorphically through 0 (value 0) and has no other zeros of  $\sin z$  inside the unit disk, so

$$\oint_{|z|=1} \frac{z^2}{\sin z} dz = 0.$$

# Real integrals by a semicircular contour

Let  $\Gamma_R$  consist of the real segment from  $-R$  to  $R$  and the upper semicircle  $C_R : z = Re^{i\theta}$ ,  $0 \leq \theta \leq \pi$ . For an integrand with no pole on the real segment,

$$\oint_{\Gamma_R} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz. \quad (1.8.1)$$

The magnitude of an integral is at most the maximum integrand magnitude  $M$  times the path length  $L$ ; this is the  $ML$  estimate. If  $M_R = \max_{z \in C_R} |f(z)|$ , the arc length is  $\pi R$ , so

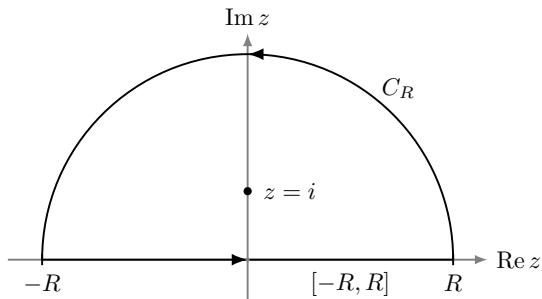
$$\left| \int_{C_R} f(z) dz \right| \leq \pi R M_R.$$

Thus the arc vanishes whenever  $RM_R \rightarrow 0$ . If the improper real integral converges, the residue theorem then gives

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k).$$

When an exponential  $e^{iaz}$  is present, its decay in one half-plane often supplies the needed estimate.

# Real integrals by a semicircular contour (cont.)



# Real integrals by a semicircular contour (cont.)

**Example:**  $\int_{-\infty}^{\infty} \cos x / (1 + x^2) dx$ . Consider

$$J = \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx.$$

For  $z = x + iy$  in the upper half-plane,

$$|e^{iz}| = |e^{ix-y}| = e^{-y} \leq 1.$$

On  $|z| = R > 1$ ,

$$|1 + z^2| \geq ||z|^2 - 1| = R^2 - 1,$$

so

$$\left| \int_{C_R} \frac{e^{iz}}{1 + z^2} dz \right| \leq \frac{\pi R}{R^2 - 1} \rightarrow 0.$$

Only the pole  $z = i$  lies in the upper half-plane. Since  $1 + z^2 = (z - i)(z + i)$ ,

$$\begin{aligned} \operatorname{Res}\left(\frac{e^{iz}}{1 + z^2}, i\right) &= \frac{e^{iz}}{z + i} \Big|_{z=i} \\ &= \frac{e^{-1}}{2i}. \end{aligned}$$

# Real integrals by a semicircular contour (cont.)

Hence

$$J = 2\pi i \frac{e^{-1}}{2i} = \frac{\pi}{e}.$$

Taking real parts gives

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}. \quad (1.8.2)$$

The imaginary part vanishes consistently because  $\sin x/(1+x^2)$  is odd.

**Example:**  $\int_{-\infty}^{\infty} dx/(1+x^4)$ . Close in the upper half-plane. The upper arc vanishes by ML:  $|\int_{C_R}| \leq \pi R/(R^4-1) \rightarrow 0$  as  $R \rightarrow \infty$ . The poles are the fourth roots of  $-1$ :

$$z_k = e^{i(\pi+2k\pi)/4}, \quad k = 0, 1, 2, 3.$$

Those in the upper half-plane are the simple poles

$$z_0 = e^{i\pi/4}, \quad z_1 = e^{i3\pi/4}.$$

Since  $1+z^4 = \prod_{k=0}^3 (z-z_k)$  and  $(1+z^4)' = 4z^3$ , each residue is

$$\text{Res}\left(\frac{1}{1+z^4}, z_j\right) = \frac{1}{4z_j^3}.$$

# Real integrals by a semicircular contour (cont.)

Since  $z_j^4 = -1$ , this residue is  $-z_j/4$ . Adding the two upper-half-plane residues,

$$\sum_{j=0}^1 \operatorname{Res}\left(\frac{1}{1+z^4}, z_j\right) = -\frac{e^{i\pi/4} + e^{3i\pi/4}}{4} = -\frac{i}{2\sqrt{2}}.$$

The residue theorem therefore gives

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}. \quad (1.8.3)$$

## Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$

Let

$$I(a) = \int_0^\infty \frac{x^a}{(1+x)^2} dx, \quad -1 < a < 1.$$

The endpoint conditions follow directly: near 0 the integrand behaves as  $x^a$ , which is integrable when  $a > -1$ ; near infinity it behaves as  $x^{a-2}$ , which is integrable when  $a < 1$ .

For  $a \neq 0$ , choose the branch of the power whose cut runs along the *positive* real axis,

$$z^a = e^{a\ell(z)}, \quad \ell(z) = \log |z| + i \arg z, \quad 0 < \arg z < 2\pi,$$

rather than the principal logarithm used earlier. The cut is chosen along the real integration interval so that the keyhole follows its two sides. Use a counterclockwise keyhole with outer radius  $R$  and inner radius  $\varepsilon$ . The hole keeps the branch point 0 outside the enclosed region. A branch point has no ordinary residue: the power is not single-valued on a full punctured disk.

On the upper side,  $z = x$  and the contribution is

$$I_{\varepsilon,R}(a) = \int_\varepsilon^R \frac{x^a}{(1+x)^2} dx.$$

## Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

On the lower side,  $z = xe^{2\pi i}$  and  $z^a = x^a e^{2\pi ia}$ . The orientation is from  $R$  back to  $\varepsilon$ , so the contribution is

$$\int_R^\varepsilon \frac{x^a e^{2\pi ia}}{(1+x)^2} dx = -e^{2\pi ia} I_{\varepsilon,R}(a).$$

Thus the two straight segments sum to

$$(1 - e^{2\pi ia}) I_{\varepsilon,R}(a).$$

We write  $F = O(G)$  when  $|F|$  is bounded by a constant times  $|G|$  in the stated limit. On the outer circle, since  $a$  is real and  $z^a = e^{a(\log R + i \arg z)}$ , one has  $|z^a| = R^a$ . Hence

$$\left| \frac{z^a}{(1+z)^2} \right| \leq \frac{R^a}{(R-1)^2} = O(R^{a-2}),$$

so the arc integral is  $O(R^{a-1}) \rightarrow 0$  because  $a < 1$ . On the inner circle,

$$\left| \frac{z^a}{(1+z)^2} \right| = O(\varepsilon^a),$$

so the arc integral is  $O(\varepsilon^{a+1}) \rightarrow 0$  because  $a > -1$ .

The only enclosed singularity is the double pole at  $z = -1$ . Since

$$(z+1)^2 \frac{z^a}{(1+z)^2} = z^a,$$

## Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

formula (1.7.3) gives

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = \frac{d}{dz} z^a \Big|_{z=-1}.$$

The chain rule gives  $(z^a)' = az^{a-1}$  on this branch. Since  $\ell(-1) = i\pi$ ,

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = ae^{i\pi(a-1)} = -ae^{i\pi a}.$$

Hence the residue theorem gives, after  $R \rightarrow \infty$  and  $\varepsilon \rightarrow 0$ ,

$$(1 - e^{2\pi ia})I(a) = -2\pi iae^{i\pi a}.$$

Factor the denominator:

$$\begin{aligned} 1 - e^{2\pi ia} &= e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) \\ &= -2ie^{i\pi a}\sin(\pi a). \end{aligned}$$

Therefore

$$\begin{aligned} I(a) &= \frac{-2\pi iae^{i\pi a}}{-2ie^{i\pi a}\sin(\pi a)} \\ &= \frac{\pi a}{\sin(\pi a)}. \end{aligned}$$

## Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

Thus

$$\int_0^\infty \frac{x^a}{(1+x)^2} dx = \frac{\pi a}{\sin(\pi a)}, \quad -1 < a < 1. \quad (1.8.4)$$

At  $a = 0$ , the right-hand side has limit 1 because  $\sin(\pi a) \sim \pi a$ , where  $u(a) \sim v(a)$  means  $u(a)/v(a) \rightarrow 1$  in the stated limit, and directly  $\int_0^\infty (1+x)^{-2} dx = 1$ .

The same keyhole with the simpler integrand  $z^{-a}/(1+z)$  (simple pole at  $-1$ ) yields the companion evaluation, valid for  $0 < a < 1$ ,

$$\int_0^\infty \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin(\pi a)}. \quad (1.8.5)$$

Here the lower-side phase is  $e^{-2\pi ia}$ , and the residue at  $-1$  is  $e^{-i\pi a}$ . The outer and inner arcs are respectively  $O(R^{-a})$  and  $O(\varepsilon^{1-a})$ , so both vanish. Therefore

$$(1 - e^{-2\pi ia}) \int_0^\infty \frac{x^{-a}}{1+x} dx = 2\pi i e^{-i\pi a}.$$

Using  $1 - e^{-2\pi ia} = 2ie^{-i\pi a} \sin(\pi a)$  gives (1.8.5). This reflection factor reappears in the Gamma and Beta functions.

# Jordan's lemma and $\int \sin x/x \, dx$

**A small clockwise semicircle around a real-axis pole.** Near a simple pole  $z_0$ , write

$$f(z) = \frac{r_{-1}}{z - z_0} + g(z), \quad r_{-1} = \text{Res}(f, z_0),$$

with  $g$  bounded. Parameterize the upper indentation by  $z = z_0 + \varepsilon e^{i\theta}$ , traversed from  $\theta = \pi$  to 0. Then

$$\frac{r_{-1}}{z - z_0} dz = ir_{-1} d\theta, \quad ir_{-1} \int_{\pi}^0 d\theta = -i\pi r_{-1}.$$

The bounded part contributes at most its bound times  $\pi\varepsilon$ , so

$$\int_{C_\varepsilon} f(z) dz \longrightarrow -i\pi \text{Res}(f, z_0). \quad (1.8.6)$$

The minus sign comes from the clockwise orientation.

## Jordan-type semicircle estimate

Let  $C_R$  be the upper semicircle and let  $M_R = \max_{z \in C_R} |f(z)|$ . If  $M_R \rightarrow 0$  as  $R \rightarrow \infty$ , then for every  $a > 0$ ,

$$\int_{C_R} f(z) e^{iaz} dz \longrightarrow 0.$$

## Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

We use this estimate without its full proof. On  $z = Re^{i\theta}$ ,  $|e^{iaz}| = e^{-aR \sin \theta}$ , and the angular integral satisfies

$$\left| \int_{C_R} f(z) e^{iaz} \, dz \right| \leq \frac{\pi M_R}{a}.$$

The estimate includes the small regions near  $\theta = 0, \pi$ , where the exponential alone does not decay. Continuity on the arc suffices for the estimate; analyticity is needed separately when applying the residue theorem.

**Example: the Dirichlet integral.** Integrate  $e^{iz}/z$  over the upper semicircle, indenting clockwise above  $z = 0$ . On the large arc, take  $f(z) = 1/z$ , so  $M_R = 1/R \rightarrow 0$  and the Jordan estimate applies. The small semicircle over  $z = 0$  is traversed clockwise through angle  $\pi$ , so by the indentation rule (1.8.6) it contributes  $-i\pi \operatorname{Res}$ :

$$\begin{aligned} -i\pi \operatorname{Res}\left(\frac{e^{iz}}{z}, 0\right) &= -i\pi \lim_{z \rightarrow 0} e^{iz} \\ &= -i\pi. \end{aligned}$$

The indentation excludes the only pole, so  $e^{iz}/z$  is holomorphic on and inside the indented contour and Cauchy's theorem makes the total integral zero. Written out,

## Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

that contour has four pieces — the segment  $[-R, -\varepsilon]$ , the small semicircle  $C_\varepsilon$  traversed clockwise from  $-\varepsilon$  to  $+\varepsilon$ , the segment  $[\varepsilon, R]$ , and the large semicircle  $C_R$ :

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{C_\varepsilon} \frac{e^{iz}}{z} dz + \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_{C_R} \frac{e^{iz}}{z} dz = 0.$$

The two real segments must be combined symmetrically because  $\cos x/x$  behaves as  $1/x$  near zero. This is the *Cauchy principal value*:

$$\text{PV} \int_{-\infty}^{\infty} g(x) dx = \lim_{\substack{\varepsilon \rightarrow 0^+ \\ R \rightarrow \infty}} \left( \int_{-R}^{-\varepsilon} g(x) dx + \int_{\varepsilon}^R g(x) dx \right).$$

The large arc tends to 0 and the small arc to  $-i\pi$ . Letting  $R \rightarrow \infty$  and  $\varepsilon \rightarrow 0$  in the four-piece identity therefore gives

$$0 = \text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx - i\pi,$$

and hence

$$\text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = i\pi. \quad (1.8.7)$$

## Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

Taking imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi. \quad (1.8.8)$$

The integrand  $\sin x/x$  is continuous at 0 after assigning the value 1, so its integral needs no principal-value interpretation there; the principal value was required only for the auxiliary exponential integral.

# Sampling kernels: writing a sum as an integral

Residue calculus also runs in the other direction: a *sum* can be written as a contour integral by multiplying the summand by a kernel whose poles sit at the sample points.

**Integer samples.** The kernel  $\pi \cot \pi z$  has a simple pole at every integer  $n$  with residue

$$\operatorname{Res}_{z=n} \pi \cot \pi z = \frac{\pi \cos \pi n}{\pi \cos \pi n} = 1.$$

For  $f$  holomorphic inside a contour  $C$  enclosing the integers  $n_1, \dots, n_2$  (and no other poles of the integrand),

$$\sum_{m=n_1}^{n_2} f(m) = \frac{1}{2\pi i} \oint_C f(z) \pi \cot \pi z \, dz.$$

The alternating variant uses  $\pi \csc \pi z = \pi / \sin \pi z$ , whose residue at  $n$  is  $(-1)^n$ , giving  $\sum (-1)^m f(m)$ .

**Half-integer samples.** The kernel  $\pi / \cos \pi z$  has residue  $(-1)^{n+1}$  at  $z = n + \frac{1}{2}$  (since  $(\cos \pi z)'|_{n+1/2} = -\pi(-1)^n$ ), so a contour enclosing the half-integers samples  $\sum (-1)^{n+1} f(n + \frac{1}{2})$ .

The payoff comes after the rewrite: deforming  $C$  moves the value of the sum onto the poles and saddle contributions of  $f$  itself. Turning a slowly converging modal

# Sampling kernels: writing a sum as an integral (cont.)

series into residues at complex poles of the summand is the Watson transformation of scattering theory; this frame is its entire complex analysis.

# The steepest-descent approximation

**Step 1: set up the large-parameter integral.** Many asymptotic formulas for special functions come from integrals of the form

$$I(s) = \int_{\Gamma} g(z) e^{sf(z)} dz, \quad s \rightarrow +\infty, \quad (1.9.1)$$

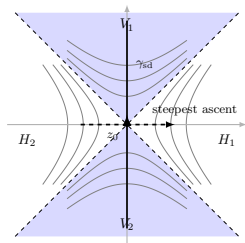
where  $f$  and  $g$  are holomorphic near a contour  $\Gamma$  that may be deformed without crossing singularities of the integrand. Write  $f = u + iv$  with real  $u, v$ . As  $s$  grows,  $e^{su}$  favors points where  $u$  is largest on the deformed contour. We study the case where a *saddle* supplies the leading contribution; endpoints must not dominate.

**Step 2: locate a nondegenerate saddle.** We seek a point  $z_0$  where

$$f'(z_0) = 0, \quad f''(z_0) \neq 0.$$

At such a point the first-order change vanishes. The quadratic term determines which directions make  $\operatorname{Re} f$  decrease. In the complex plane these are descent directions through a saddle, rather than a maximum in every direction.

# The steepest-descent approximation (cont.)



In the schematic, hill sectors and valley sectors meet at the saddle; a steepest-descent path through the saddle runs into the valleys of  $u = \operatorname{Re} f$ .

# The steepest-descent approximation (cont.)

**Step 3: choose the descent direction.** Taylor expansion at a nondegenerate saddle is

$$f(z) = f(z_0) + \frac{1}{2}f''(z_0)(z - z_0)^2 + O((z - z_0)^3). \quad (1.9.2)$$

Write

$$f''(z_0) = |f''(z_0)|e^{i\theta}, \quad z - z_0 = re^{i\phi}.$$

Then

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = \frac{1}{2}|f''(z_0)|r^2 e^{i(\theta+2\phi)}.$$

A *steepest-descent* direction makes this quadratic change negative real, so

$$\theta + 2\phi = \pi \pmod{2\pi}, \quad \text{that is} \quad \phi = \frac{\pi - \theta}{2} \pmod{\pi}. \quad (1.9.3)$$

The ambiguity by  $\pi$  is not a defect: it is the two opposite rays leaving  $z_0$ , and Step 4 uses both at once by letting  $r$  run over all of  $\mathbb{R}$ . Along that ray,

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = -\frac{1}{2}|f''(z_0)|r^2.$$

(The orthogonal choice  $\theta + 2\phi = 0$  is steepest *ascent*.)

# The steepest-descent approximation (cont.)

**Step 4: reduce to a Gaussian integral.** Assume the contour can be deformed through  $z_0$  along both descent rays, that this saddle dominates the integral, and that  $g(z_0) \neq 0$ . Parameterize the full descent line by

$$\alpha = e^{i(\pi-\theta)/2} = i e^{-i\theta/2}, \quad z = z_0 + \alpha r, \quad r \in \mathbb{R},$$

Thus  $dz = \alpha dr$ . Since  $\alpha^2 = -e^{-i\theta}$ , we have  $f''(z_0)\alpha^2 = -|f''(z_0)|$ . Replacing  $f$  by its quadratic Taylor piece and  $g$  by the constant  $g(z_0)$  yields

$$\begin{aligned} I(s) &\sim g(z_0) e^{sf(z_0)} \alpha \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}s|f''(z_0)|r^2\right) dr \\ &= g(z_0) e^{sf(z_0)} \alpha \sqrt{\frac{2\pi}{s|f''(z_0)|}}, \end{aligned}$$

where we used  $\int_{-\infty}^{\infty} e^{-ar^2} dr = \sqrt{\pi/a}$  with  $a = \frac{1}{2}s|f''(z_0)|$ . This standard Gaussian integral is evaluated in Chapter 2.

Reverse orientation changes the sign; one descent ray gives half the Gaussian. For several equally dominant saddles, add their contributions. Therefore, for the stated orientation,

$$I(s) \sim g(z_0) e^{sf(z_0)} i e^{-i\theta/2} \sqrt{\frac{2\pi}{s|f''(z_0)|}}. \quad (1.9.4)$$

# Real special cases: Laplace method and stationary phase

The same local-Gaussian idea appears in two real-variable forms used later in the course.

**Laplace's method (real, decaying exponential).** For a smooth real phase  $\varphi$  with a unique interior maximum at  $x_0$ ,  $\varphi'(x_0) = 0$ ,  $\varphi''(x_0) < 0$ , and a smooth amplitude  $g$  with  $g(x_0) \neq 0$ ,

$$\int_a^b g(x) e^{s\varphi(x)} dx \sim g(x_0) e^{s\varphi(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty. \quad (1.9.5)$$

This is (1.9.4) restricted to the real line with  $\theta = \pi$  (so  $f''(x_0) = -|\varphi''(x_0)|$  is already negative real and  $\alpha = 1$ ). Chapter 2 uses this pattern for Stirling's approximation of  $n!$  from the gamma integral.

**Stationary phase (real, oscillatory).** For a real phase  $\varphi$  with  $\varphi'(x_0) = 0$ ,  $\varphi''(x_0) \neq 0$ ,

$$\int g(x) e^{is\varphi(x)} dx \sim g(x_0) e^{is\varphi(x_0)} e^{i(\pi/4) \operatorname{sgn} \varphi''(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty, \quad (1.9.6)$$

Here  $\operatorname{sgn} q$  is  $+1$  for  $q > 0$  and  $-1$  for  $q < 0$ . Assume the smooth amplitude is zero near the endpoints,  $x_0$  is the only stationary point where the amplitude is nonzero,

# Real special cases: Laplace method and stationary phase (cont.)

and  $g(x_0) \neq 0$ . Contributions from several nondegenerate stationary points are added. The Fresnel phase factor arises from the local Gaussian limit. Near  $x_0$ ,

$$\varphi(x) \approx \varphi(x_0) + \frac{1}{2}\varphi''(x_0)(x - x_0)^2.$$

Set  $u = \sqrt{s}(x - x_0)$ , so  $dx = du/\sqrt{s}$ . Then

$$\int g(x) e^{is\varphi(x)} dx \sim e^{is\varphi(x_0)} \frac{g(x_0)}{\sqrt{s}} \int_{-\infty}^{\infty} e^{i(\varphi''(x_0)/2)u^2} du.$$

The oscillatory Gaussian has the corresponding formula

$$\int_{-\infty}^{\infty} e^{\pm iau^2} du = \sqrt{\frac{\pi}{a}} e^{\pm i\pi/4}, \quad a > 0.$$

One way to obtain it is to insert a damping factor  $e^{-\varepsilon u^2}$ , use  $\sqrt{\pi/(\varepsilon \mp ia)}$ , and let  $\varepsilon \rightarrow 0^+$ . We use the resulting Fresnel formula without its convergence proof. The positive and negative curvatures give the factors  $e^{i\pi/4}$  and  $e^{-i\pi/4}$ , respectively: together they give (1.9.6). Chapter 5 applies this rule to the unit-circle integral for  $J_n(x)$ .

# Real special cases: Laplace method and stationary phase (cont.)

**Two refinements to carry forward.** When the dominant point is an *endpoint* rather than an interior saddle or stationary point, repeated integration by parts replaces the Gaussian. The simplest case is *Watson's lemma*: for smooth  $g$  with sufficient decay,

$$\int_0^\infty g(t)e^{-st} dt \sim \frac{g(0)}{s} + \frac{g'(0)}{s^2} + \cdots, \quad s \rightarrow +\infty.$$

The endpoint supplies algebraic, not  $1/\sqrt{s}$ , decay.

Second, (1.9.4) assumes the amplitude  $g$  varies slowly near the saddle. If a pole of the integrand approaches or crosses the saddle—the geometry behind a transition region in wave problems—the single-Gaussian answer is no longer uniform, and an error-function-type transition function takes its place. Recognizing which refinement a problem needs is part of using the method.

## Worked example: large binomial coefficient

Let  $m, n$  be integers with  $0 < m < n$ . As  $n \rightarrow \infty$ , keep  $t = m/n$  away from 0 and 1. From  $(1+z)^n = \sum_{k=0}^n \binom{n}{k} z^k$ , Cauchy's coefficient formula gives

$$\binom{n}{m} = \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}} dz = \oint g(z) e^{nf(z)} dz,$$

where

$$g(z) = \frac{1}{2\pi i z}, \quad f(z) = \log(1+z) - t \log z$$

(with branches chosen locally near the positive saddle below).

**Saddle.** Differentiate:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \quad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Set  $f'(z_0) = 0$ :

$$z_0 - t(1+z_0) = 0 \implies z_0 = \frac{t}{1-t} = \frac{m}{n-m} > 0.$$

## Worked example: large binomial coefficient (cont.)

Then  $1 + z_0 = 1/(1 - t)$ , and

$$\begin{aligned}f(z_0) &= \log \frac{1}{1-t} - t \log \frac{t}{1-t} = -t \log t - (1-t) \log(1-t), \\f''(z_0) &= -\frac{1}{(1+z_0)^2} + \frac{t}{z_0^2} = -(1-t)^2 + \frac{(1-t)^2}{t} \\&= (1-t)^2 \left( \frac{1}{t} - 1 \right) = \frac{(1-t)^3}{t} > 0.\end{aligned}$$

Thus  $\theta = \arg f''(z_0) = 0$ , so the descent direction from (1.9.3) is  $\phi = \pi/2$ : the local descent line is vertical. Take the contour in Cauchy's coefficient formula to be the circle  $|z| = z_0$ . On this circle,  $|z|$  is fixed and

$$|1 + z_0 e^{i\psi}|^2 = 1 + 2z_0 \cos \psi + z_0^2$$

is largest at  $\psi = 0$ . The saddle  $z = z_0$  therefore dominates. The counterclockwise tangent there points upward ( $+i$ ), matching the descent direction and fixing the sign of the result.

**Apply (1.9.4) with  $s = n$ .**

$$\binom{n}{m} \sim \frac{1}{2\pi i z_0} e^{nf(z_0)} i \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1}{2\pi z_0} e^{nf(z_0)} \sqrt{\frac{2\pi}{nf''(z_0)}}.$$

## Worked example: large binomial coefficient (cont.)

Substitute  $1/z_0 = (1-t)/t$  and  $1/f''(z_0) = t/(1-t)^3$ :

$$\frac{1}{2\pi z_0} \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1-t}{2\pi t} \sqrt{\frac{2\pi t}{n(1-t)^3}} = \frac{1}{\sqrt{2\pi nt(1-t)}}.$$

Also  $e^{nf(z_0)} = \exp[-n(t \log t + (1-t) \log(1-t))]$ , so

$$\binom{n}{m} \sim \frac{\exp[-n(t \log t + (1-t) \log(1-t))]}{\sqrt{2\pi nt(1-t)}}. \quad (1.9.7)$$

Chapter 2's Stirling formula gives the same leading approximation from the factorial expression for  $\binom{n}{m}$ .

# Summary

- Complex differentiability requires one direction-independent difference-quotient limit. With continuous first partial derivatives, this is equivalent to the Cauchy–Riemann equations  $u_x = v_y$ ,  $u_y = -v_x$ .
- Cauchy's theorem implies path independence on simply connected domains. Cauchy's integral formula yields all derivatives and the local Taylor expansion.
- A Laurent series is valid on an annulus. Its principal part classifies an isolated singularity as removable, a pole, or essential. Branch points are not isolated and do not possess ordinary residues.
- The residue is the coefficient  $a_{-1}$ . The residue theorem evaluates a closed contour integral as  $2\pi i$  times the sum of enclosed residues.
- Three real-integral templates: semicircle (rational/exponential), keyhole (powers and branch cuts, as in  $\int_0^\infty x^a/(1+x)^2 dx$ ), and Jordan's indented semicircle (oscillatory integrals such as  $\int \sin x/x dx$ ). Always check orientation and vanishing arcs before taking limits.
- **Steepest descent** (1.9.4) is a local Gaussian approximation at a nondegenerate dominant saddle. Real special cases: Laplace method (1.9.5) (Stirling in Ch. 2) and stationary phase (1.9.6) (large- $x$  Bessel in Ch. 5). Orientation and the phase of the square root are part of the answer.